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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 03/16/2007 |
| **PI Name:** | Hlatky, Lynn Ph.D. | | |
| **Project Title:** | Computational Modeling of Chromosome Aberrations Produced by HZE Particles | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 02135-2907 | **Congressional District:** | 8 |
| **Comments:** | PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Space Radiation Biology and Shielding 03-OBPR-07 |
| **Start Date:** | 05/15/2004 | **End Date:** | 06/01/2008 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Hahnfeldt, Philip ( Caritas St. Elizabeth's Medical Center, Tufts Univ School of Medicine )<br>Sachs, Rainer ( University of California ) | | |
| **Grant/Contract No.:** | NNJ06HA27G ; NNJ04HF42G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Chromosome aberrations produced by high LET radiations are directly and indirectly relevant to space exploration and risk estimation in a number of different ways. They are important as probes for DNA repair and recombination mechanisms, as biomarkers of dose and radiation quality, as factors in genomic instability and carcinogenesis, and as surrogates for cancer in risk estimation. Quantitative modeling is needed to exploit and augment the rapidly accumulating empirical data on aberration types and frequencies. This project will apply mathematical and computer algorithms to data on aberrations produced by iron ions (56Fe). The analysis will be compared to analyses of damage produced by photons, electrons, or alpha particles, with the goal of assessing signature responses to 56Fe, an important radiation encountered in space. To perform such studies, the team has tied damage endpoints to risk through our customized software and algorithms. These endpoints include: 1. aberration complexity, quantified by cycle structures; 2. apparently and truly incomplete aberrations, related to segment size distributions; 3. aberration spectra as related to chromosome geometry, to DNA double strand break (DSB) clustering depending on radiation track structure, and especially to DSB misrejoining depending on proximity; 4. mFISH color junctions for each specific chromosome pair, related to chromosome-chromosome spatial correlations in an interphase nucleus.<br>Substantial preliminary studies have recently been carried out by our team for each aim. As remarked, we will employ the team's specialized and comprehensive software, including the highly-developed chromosome aberration simulator (CAS) software, for analyzing the aberration spectra encountered. The data to be analyzed will be for human cells in vitro, scored by mFISH or other similar techniques. A number of these data sets are being made available by many of our colleagues worldwide. This will greatly expand the range of suitable data available for analysis. We have also recently devised methods for systematically comparing mFISH data with data obtained by older methods such as FISH or solid staining and with newer methods such as color banding.<br><br>For Specific Aim 1, we will replace the current approach of using "obligate" (deterministic) cycles with a more accurate computer-based approach involving cycle probabilities. For Aim 2, new methods are being developed that allow systematic comparison of apparently incomplete aberrations with corresponding complete aberrations and corresponding truly incomplete ones, even in the case of very complex aberrations. Under Aim 2 we will also consider size distributions of fragments that have one telomere and one unrejoined free end, for comparison to recent data using pan-telomeric probes; and analyze mixtures of different radiations. Aim 3 will emphasize more realistic computer algorithms for break misrejoining, bringing this aspect of aberration modeling more nearly in line with the very sophisticated models now being used by various groups for assessing the influence of track structure and chromatin geometry on DSB locations. For Aim 4 we plan to apply a method we previously used for low LET data to data for high LET, including iron ions. The method involves testing frequencies of color junction pairs for deviations from randomness. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Improved biodosimetry |
| **Task Progress:** | Grant 03-OBPR-07. Computational Modeling of Chromosome Aberrations Produced by HZE Particles. Lynn Hlatky PI, Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine.<br>Progress for the reporting period, 05/16/2006-05/16/2007.<br><br>Under aim 1 we did, as planned, compute complexity for iron ion MFISH data (kindly supplied by M. Cornforth). It was found that the complexity was much larger than for photons or even alpha particles, and was nearly dose independent, corresponding to 1-track action. That is, the complex aberrations come mainly from double strand break interactions among double strand breaks made by a single track. The results were reported in a peer-reviewed paper in Press, Levy et al. (2007), reported in the bibliography.<br><br>We continued to make further progress on Aim 2 which deals with systematically comparing apparently incomplete, truly incomplete, and complete aberrations. In brief, the optimal procedure is to identify incompleteness type by looking for mismatches between centromeres and apparent telomeres involved in a particular aberration since in a complete aberration there are two telomeres of each color for each centromere of that color. Once incompleteness type has been characterized, color-by-color, adding "virtual fragments" of appropriate colors gives a fully methodical way to see what the possible completions are. During this reporting period we found that true incompleteness, where chromosome fragments fail to rejoin or misrejoin, can be systematically analyzed with corresponding apparently incomplete aberrations, by allowing each virtual fragment to either misrejoin or fail to rejoin. An applet developed during preliminary studies, is already available on the internet at the URL <a target="_blank" href="http://radiobiology.math.berkeley.edu/CAA/index.html">http://radiobiology.math.berkeley.edu/</a> . It has been updated with new algorithms that give a more realistic description of the probabilities for different complete aberrations that can correspond to a given apparently incomplete mFISH aberration; previously, apparent incompleteness has been a major confounding factor in analyzing HZE data on aberrations. These results and their application to iron ion data are reported in the Levy et al. paper (2007).<br><br>We continue to make progress on Aim 3, on break clustering and aberration formation. Another aspect of Aim 3 was DSB (DNA double-strand break) clustering at high LET and its influence on misrejoining frequencies. We have recently collaborated with A. Ponomarev and F. Cucinotta of NASA in work on how to subtract out background values of DNA fragment size distributions in PFGE experiments on radiation breakage to analyze DSB yields and clustering. The group has found a fully general subtraction method, based on stochastic point-process theory, which works for non-random (high LET) double strand break induction as well as random (low LET) double strand break induction and for random or non-random background breakage. The method is also applicable to subtracting out low LET effects on break distributions in the case of mixed HZE radiations and protons on DSB clustering. The results are reported in two peer reviewed papers Ponomarev et al. (2006) and (2007).<br><br>As regards aim 4, we did carry out proximity calculations for aberrations produced by high LET radiation, finding that, contrary to our expectations, such radiation is no more effective at uncovering chromosome-chromosome statistical associations during cell cycle interphase than are x-rays.<br><br>Cumulative bibliography: |

1) Full-Color Painting Reveals an Excess of Radiation-Induced Dicentrics Involving Homologous Chromosomes. Y. Plan, L. Hlatky, P. Hahnfeldt, R. Sachs, B. Loucas, and M. Cornforth. Int J Radiat Biol 2005; 81:613-20.

2) Levy D, L Hlatky and RK Sachs (2006), Software for Cycle Analysis, <a target="_blank" href="http://radiobiology.berkeley.edu/CAA/">http://radiobiology.berkeley.edu/</a> . NSF

3) SCHIP: Statistics for Chromosome Interphase Positioning. Vives S, Loucas B, Vazquez M, Brenner DJ, Sachs RK, Hlatky L, Cornforth M, Arsuaga J. Bioinformatics. 2005 21:3181-2.

4) Ponomarev, A. L., Belli, M., Hahnfeldt, P. J., Hlatky, L., Sachs, R. K., and Cucinotta, F. A., 2006, A robust procedure for removing background damage in assays of radiation-induced DNA fragment distributions, Radiat Res 166(6):908-16.

5) In press, Journal of Computational Biology: Interpreting Chromosome Aberration Spectra. Dan Levy, Chris Reeder, Bradford Loucas, Lynn Hlatky, Allen Chen, Michael Cornforth, Rainer Sachs.

6) In press 2007. Rad. Environ. Biophysics. Subtraction of Background Damage in PFGE Experiments on DNA Fragment-Size Distributions. A.L. Ponomarev, L. Hlatky, R.K. Sachs, P. Hahnfeldt, F.A. Cucinotta.

7) Shuryak, I., Sachs, R. K., Hlatky, L., Little, M. P., Hahnfeldt, P., and Brenner, D. J., 2006, Radiation-induced leukemia at doses relevant to radiation therapy: modeling mechanisms and estimating risks, J Natl Cancer Inst 98(24):1794-806.

8) Human X chromosome reveals pronounced early population subdivisions and recent bottlenecks Allen M Chen1, Jack S Chen2, Lynn Hlatky3, Caoan Wang4, Dan Levy1, Mariel Vazquez1,5, Rainer K Sachs1. Submitted to Nature

| **Bibliography Type:** | Description: (Last Updated: 11/01/2023) |
|---|---|
| **Articles in Peer-reviewed Journals** | Ponomarev AL, Belli M, Hahnfeldt PJ, Hlatky L, Sachs RK, Cucinotta FA. "A robust procedure for removing background damage in assays of radiation-induced DNA fragment distributions." Radiat Res. 2006 Dec;166(6):908-16. PMID: 17149980 , Dec-2006 |
| **Articles in Peer-reviewed Journals** | Levy D, Reeder C, Loucas B, Hlatky L, Chen A, Cornforth M, Sachs R. "Interpreting Chromosome Aberration Spectra. " Journal of Computational Biology, In Press, March 2007. , Mar-2007 |
| **Articles in Peer-reviewed Journals** | Shuryak I, Sachs RK, Hlatky L, Little MP, Hahnfeldt P, Brenner DJ. "Radiation-induced leukemia at doses relevant to radiation therapy: modeling mechanisms and estimating risks." J Natl Cancer Inst. 2006 Dec 20;98(24):1794-806. PMID: 17179481 , Dec-2006 |
| **Articles in Peer-reviewed Journals** | Ponomarev AL, Hlatky L, Sachs RK, Hahnfedlt, P, Cucinotta FA. "Subtraction of Background Damage in PFGE Experiments on DNA Fragment-Size Distributions." Radiation and Environmental Biophysics, in press, May 2007. , May-2007 |